Exemple d'utilisation du package levysim

Sommaire

Préparation **1**

I - Une trajectoire **1**
Cas 1 : inspections régulières . . . 2
Cas 2 : inspections croissantes . . . 5
Cas 3 : inspections décroissantes . . . 8
Cas 4 : inspections mixtes . . . 11

II - Plusieurs trajectoires **14**
Cas 1 : inspections régulières . . . 14
Cas 2 : inspections croissantes . . . 17
Cas 3 : inspections décroissantes . . . 20
Cas 4 : inspections mixtes . . . 23

III - Récupérer les données simulées **26**

Préparation

Installation du package depuis GitHub :

```
remotes::install_github("rytracetito/levy_packages", subdir = "levysim")
```

Tous les exemples utilisent les **processus homogènes** : les valeurs par défaut (`A`, `Lambda` et `tau` égales à l'identité).

Paramètres utilisés :

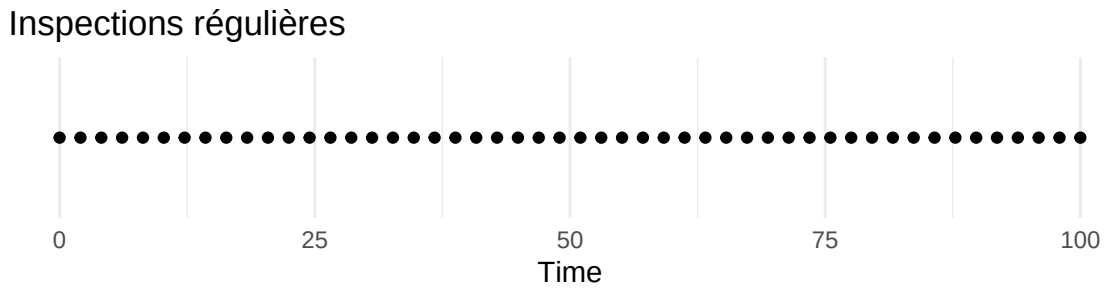
Processus	Fonction	Paramètres
Gamma	`sim_gamma`	`a = 2`, `b = 3` (taux)
Wiener	`sim_wiener`	`mu = 0.5`, `sigma = 1.5`
Inverse gaussien	`sim_ig`	`mu = 0.5`, `eta = 2`
Variance-gamma	`sim_vg`	`theta = 0.5`, `sigma = 1.5`, `nu = 2`
Meixner	`sim_meixner`	`a = 1`, `b = 0.3`, `d = 1.5`, `m = 0.2`

I - Une trajectoire

Une trajectoire inspectée 50 fois entre 0 et 100.

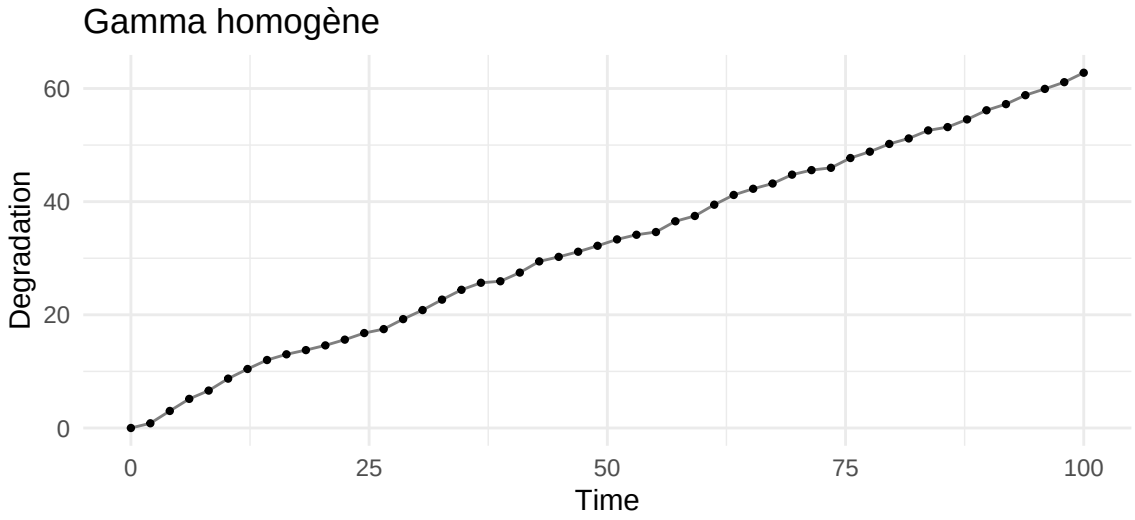
Cas 1 : inspections régulières

```
set.seed(1)
t1 <- inspection_times("regular", n = 49, tmax = 100)$time
plot_plan(t1, "Inspections régulières")
```



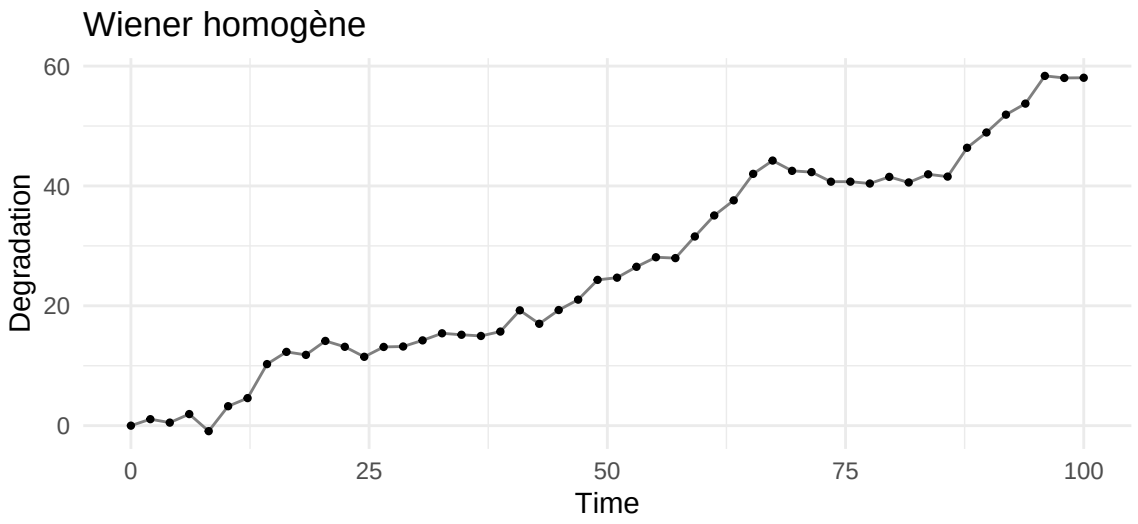
Modèle 1 : gamma homogène

```
plot_traj(sim_gamma(t1, a = 2, b = 3), "Gamma homogène")
```



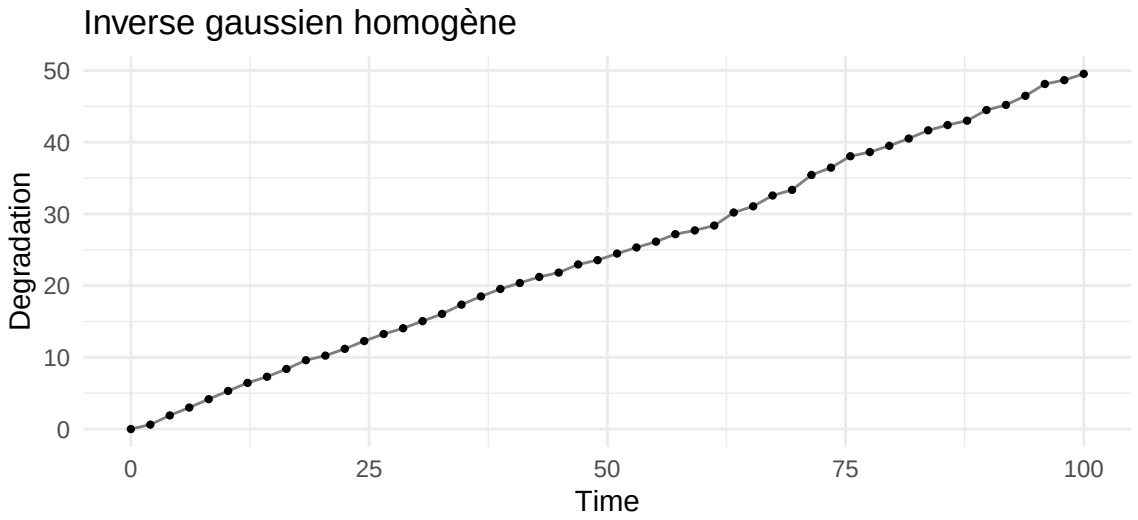
Modèle 2 : Wiener homogène

```
plot_traj(sim_wiener(t1, mu = 0.5, sigma = 1.5), "Wiener homogène")
```



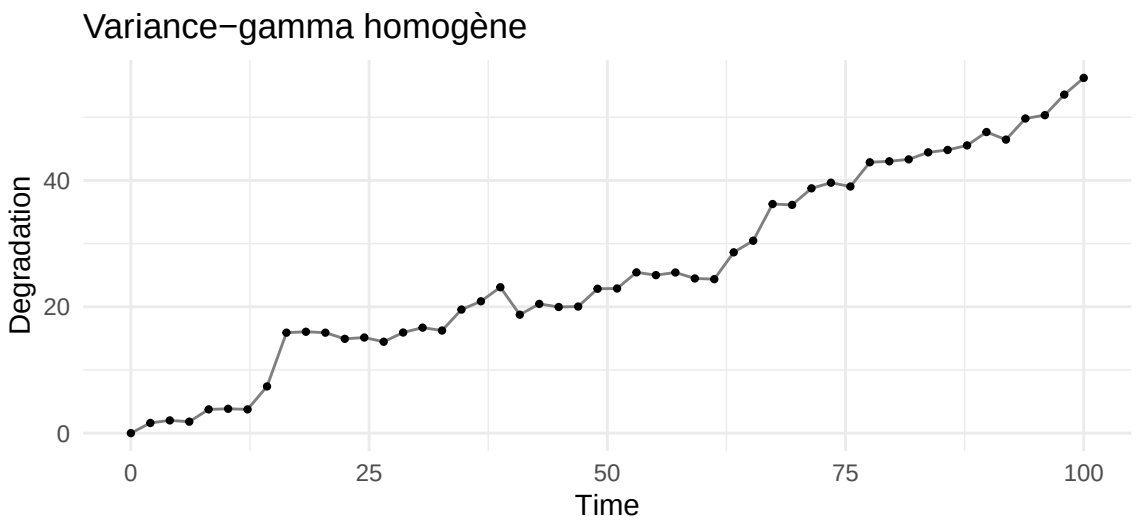
Modèle 3 : inverse gaussien

```
plot_traj(sim_ig(t1, mu = 0.5, eta = 2), "Inverse gaussien homogène")
```



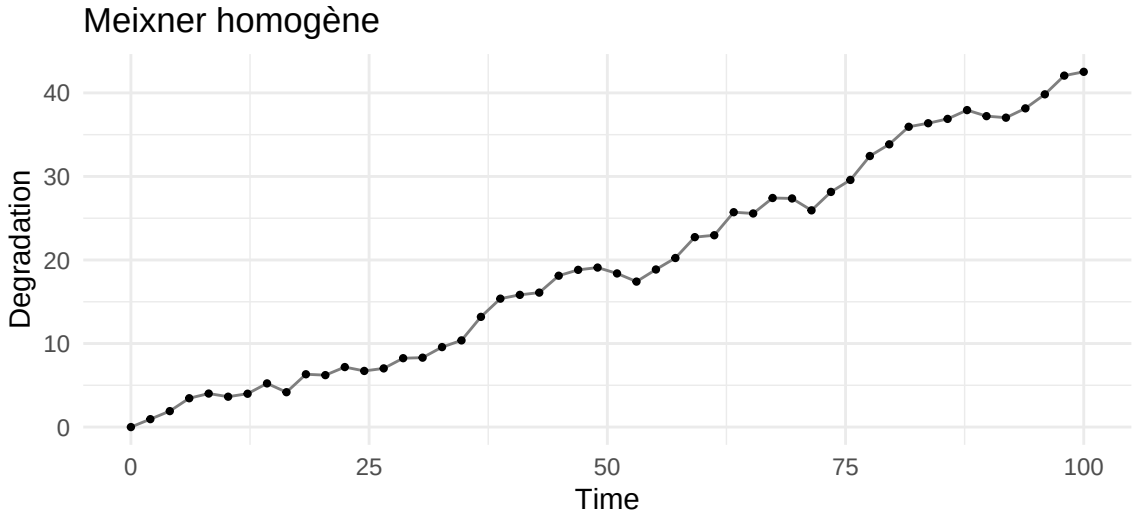
Modèle 4 : variance-gamma

```
plot_traj(sim_vg(t1, theta = 0.5, sigma = 1.5, nu = 2), "Variance-gamma homogène")
```



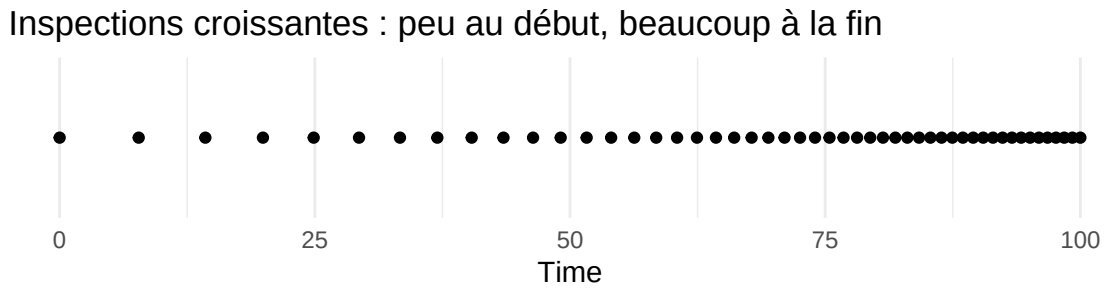
Modèle 5 : Meixner

```
plot_traj(sim_meixner(t1, a = 1, b = 0.3, d = 1.5, m = 0.2), "Meixner homogène")
```



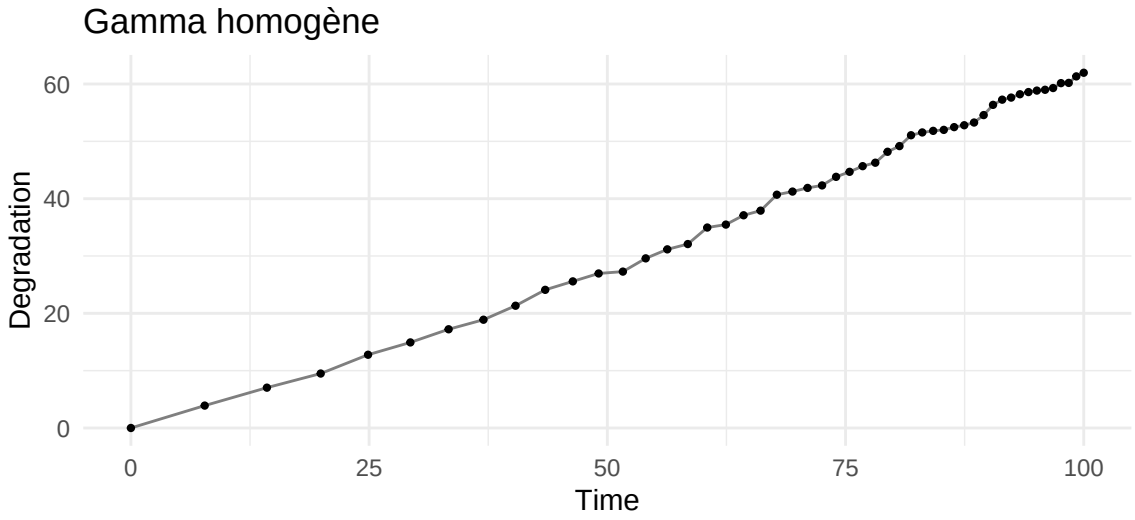
Cas 2 : inspections croissantes

```
set.seed(2)
t2 <- inspection_times("increasing", n = 49, tmax = 100)$time
plot_plan(t2, "Inspections croissantes : peu au début, beaucoup à la fin")
```



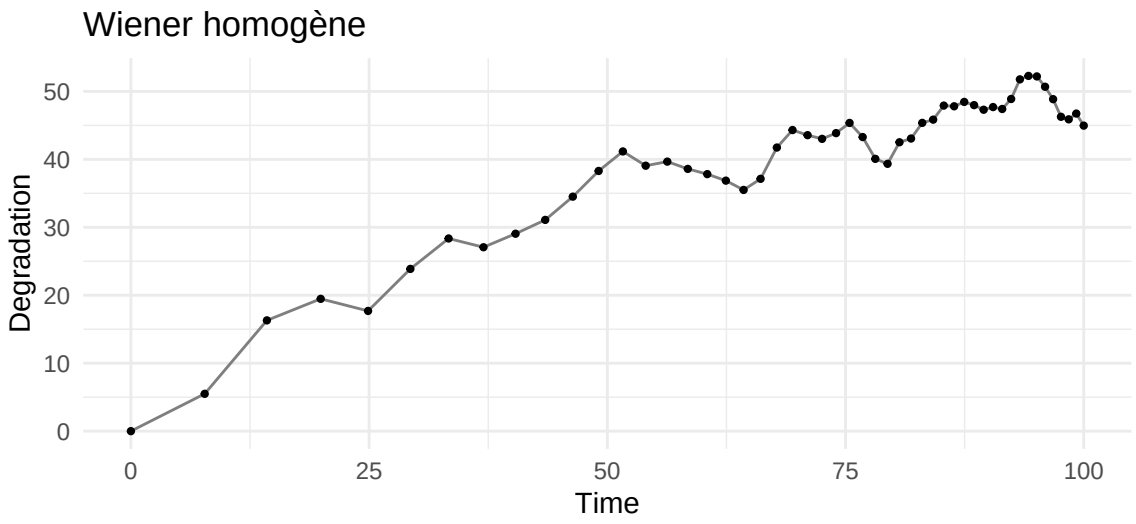
Modèle 1 : gamma homogène

```
plot_traj(sim_gamma(t2, a = 2, b = 3), "Gamma homogène")
```



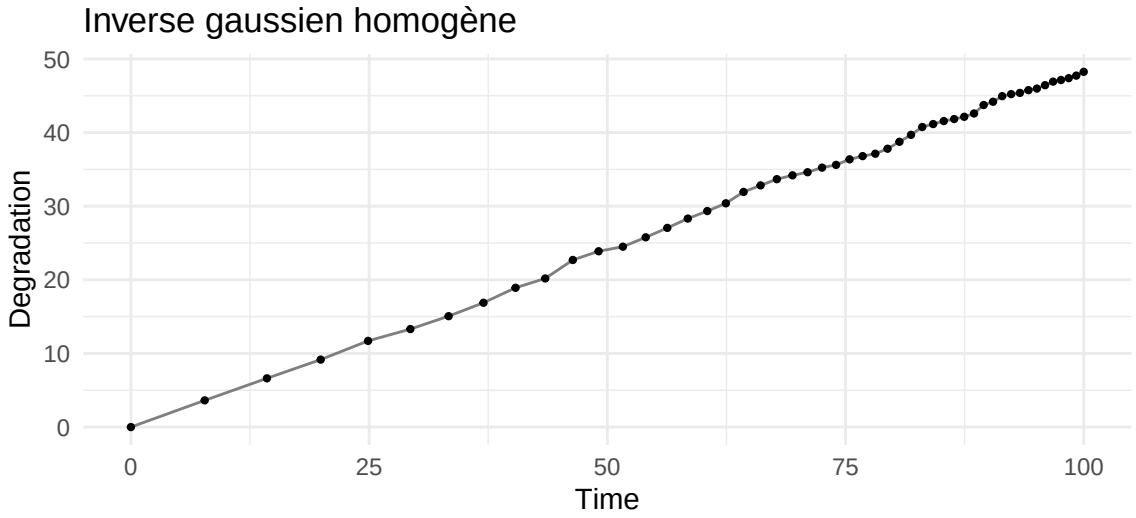
Modèle 2 : Wiener homogène

```
plot_traj(sim_wiener(t2, mu = 0.5, sigma = 1.5), "Wiener homogène")
```



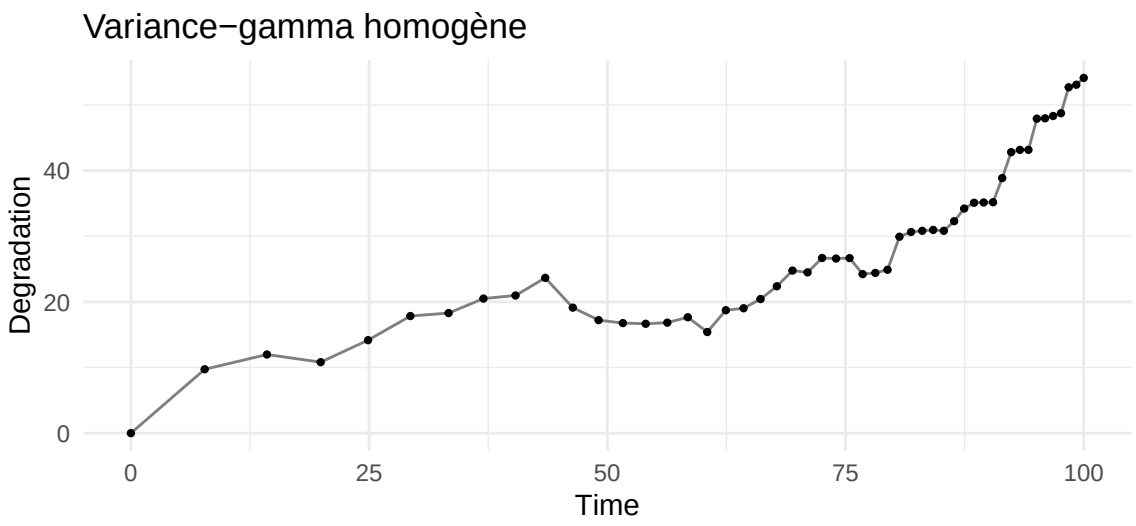
Modèle 3 : inverse gaussien

```
plot_traj(sim_ig(t2, mu = 0.5, eta = 2), "Inverse gaussien homogène")
```



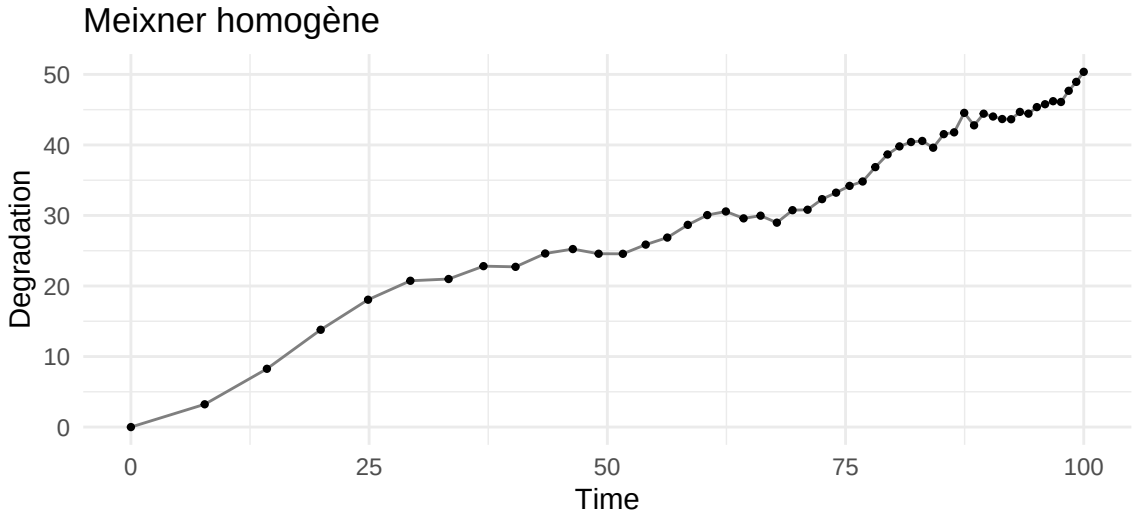
Modèle 4 : variance-gamma

```
plot_traj(sim_vg(t2, theta = 0.5, sigma = 1.5, nu = 2), "Variance-gamma homogène")
```



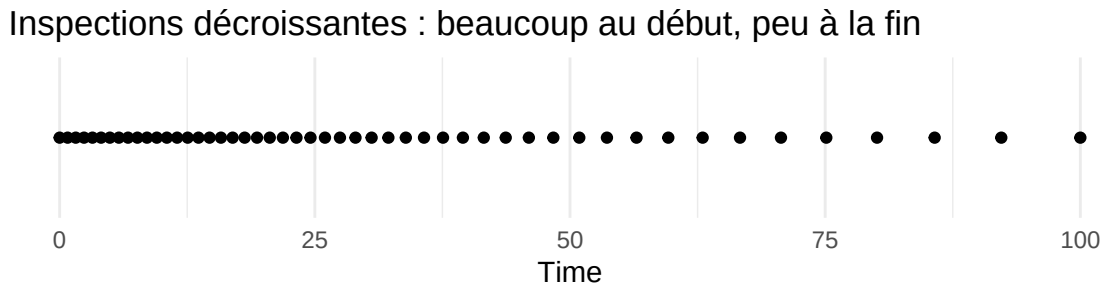
Modèle 5 : Meixner

```
plot_traj(sim_meixner(t2, a = 1, b = 0.3, d = 1.5, m = 0.2), "Meixner homogène")
```



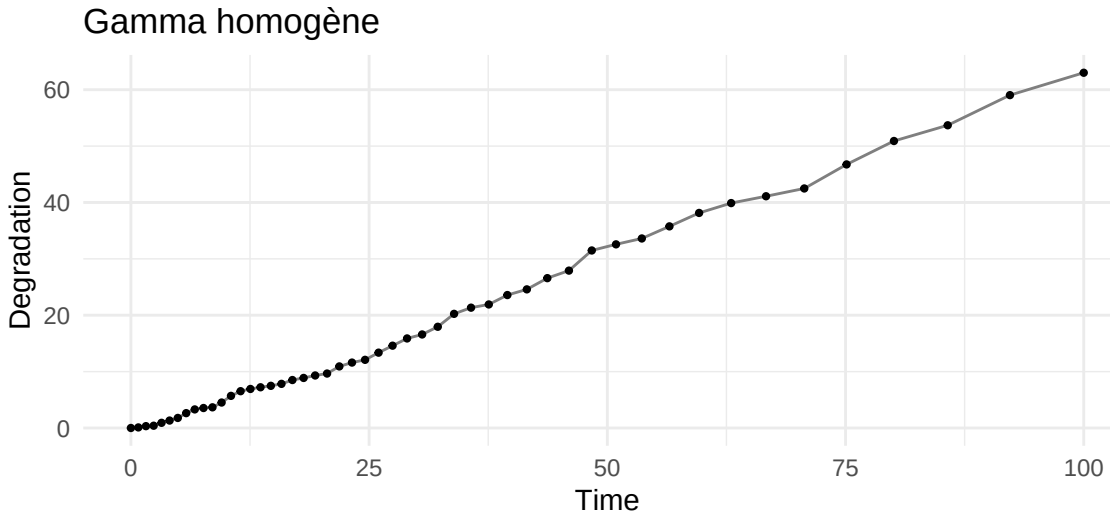
Cas 3 : inspections décroissantes

```
set.seed(3)
t3 <- inspection_times("decreasing", n = 49, tmax = 100)$time
plot_plan(t3, "Inspections décroissantes : beaucoup au début, peu à la fin")
```



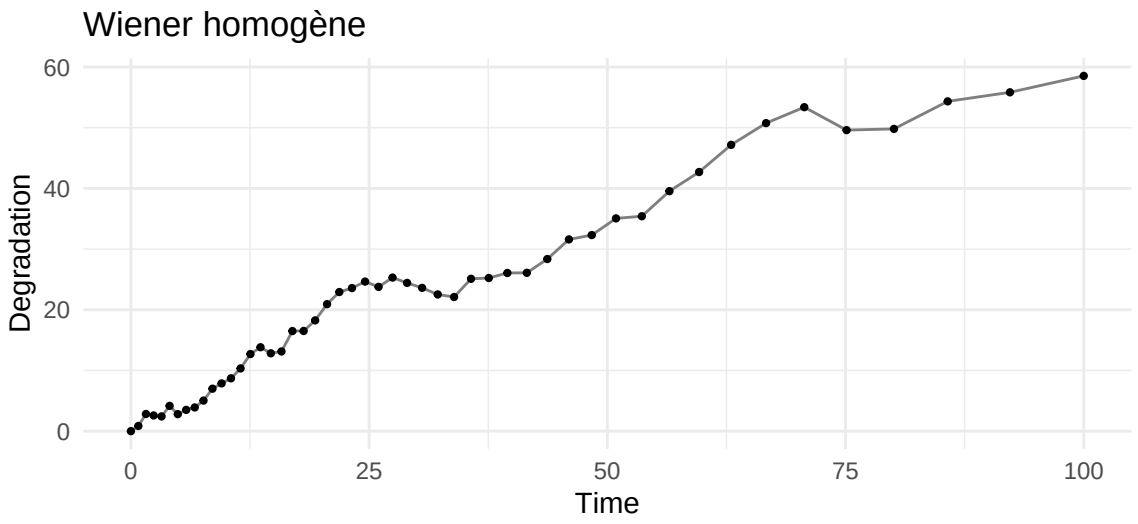
Modèle 1 : gamma homogène

```
plot_traj(sim_gamma(t3, a = 2, b = 3), "Gamma homogène")
```



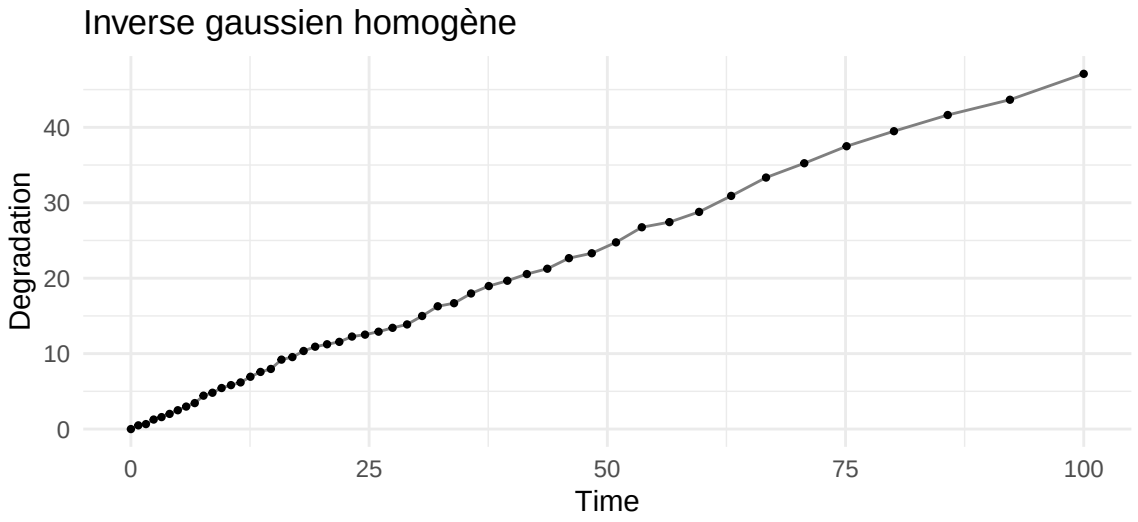
Modèle 2 : Wiener homogène

```
plot_traj(sim_wiener(t3, mu = 0.5, sigma = 1.5), "Wiener homogène")
```



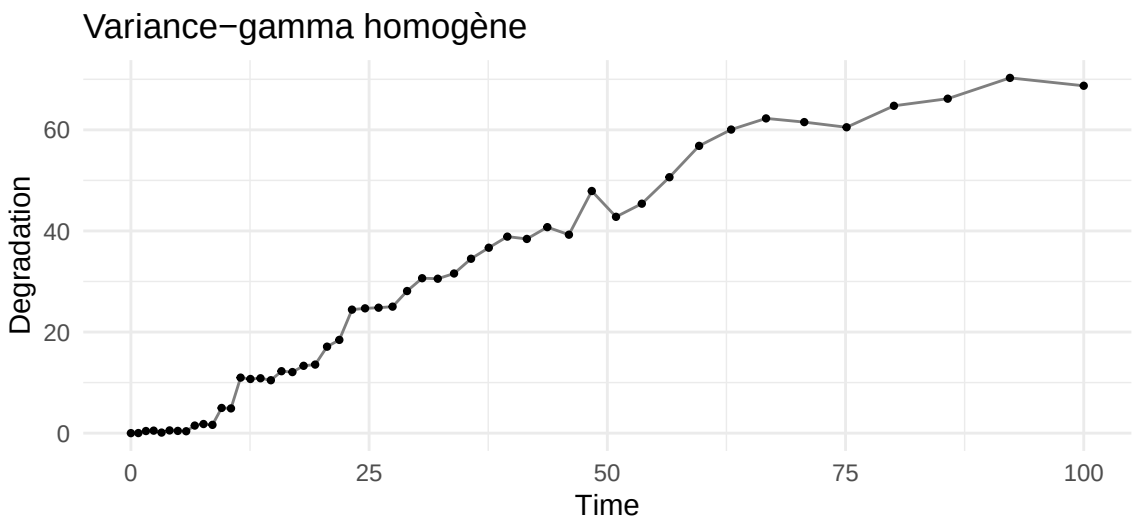
Modèle 3 : inverse gaussien

```
plot_traj(sim_ig(t3, mu = 0.5, eta = 2), "Inverse gaussien homogène")
```



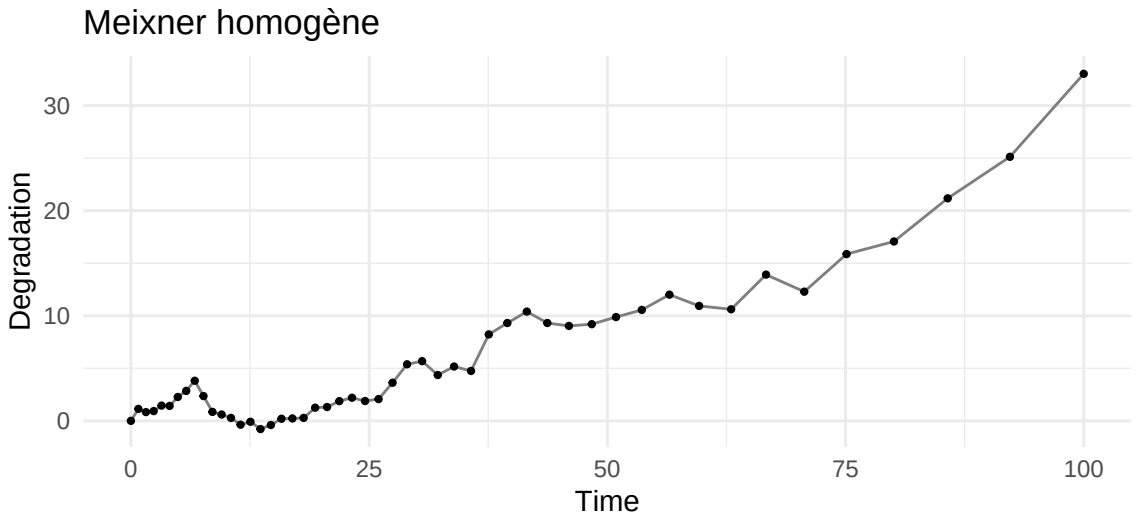
Modèle 4 : variance-gamma

```
plot_traj(sim_vg(t3, theta = 0.5, sigma = 1.5, nu = 2), "Variance-gamma homogène")
```



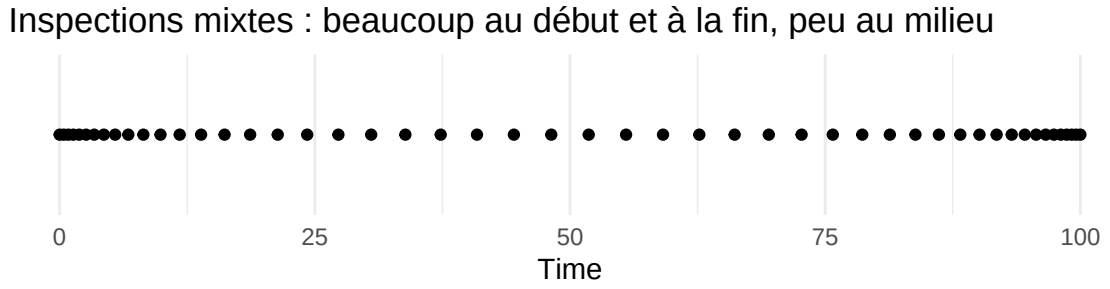
Modèle 5 : Meixner

```
plot_traj(sim_meixner(t3, a = 1, b = 0.3, d = 1.5, m = 0.2), "Meixner homogène")
```



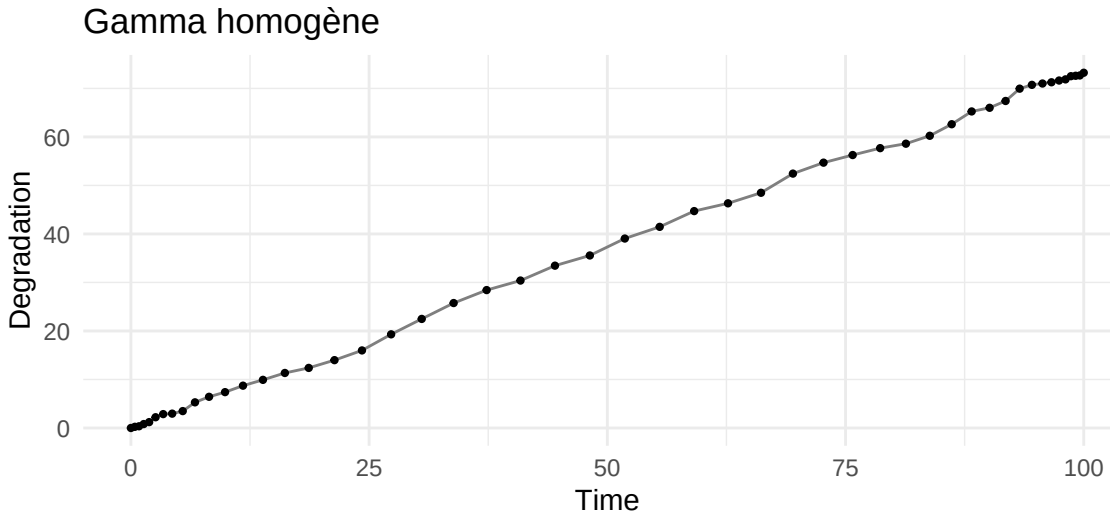
Cas 4 : inspections mixtes

```
set.seed(4)
t4 <- inspection_times("dec-inc", n = 49, tmax = 100)$time
plot_plan(t4, "Inspections mixtes : beaucoup au début et à la fin, peu au milieu")
```



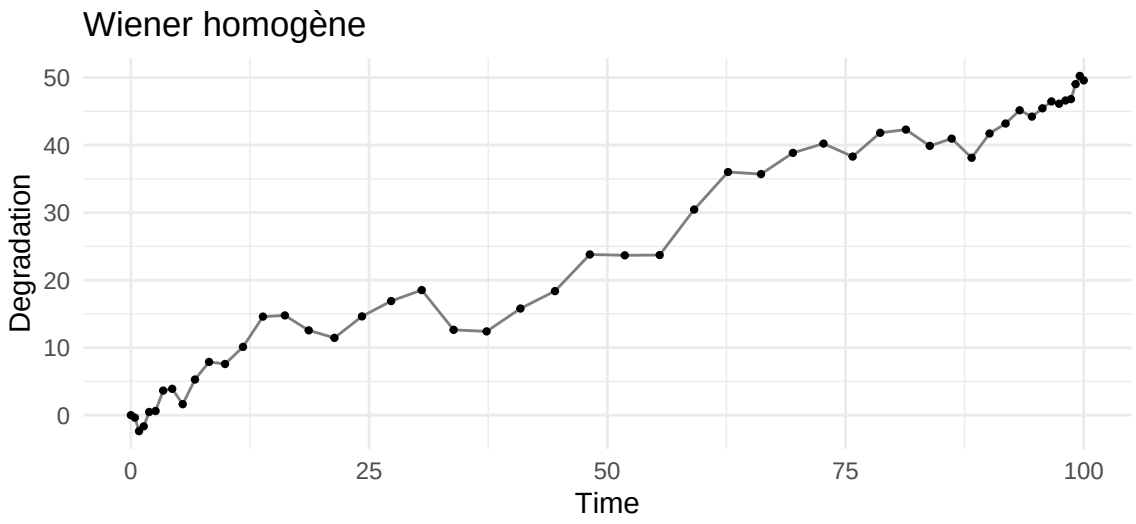
Modèle 1 : gamma homogène

```
plot_traj(sim_gamma(t4, a = 2, b = 3), "Gamma homogène")
```



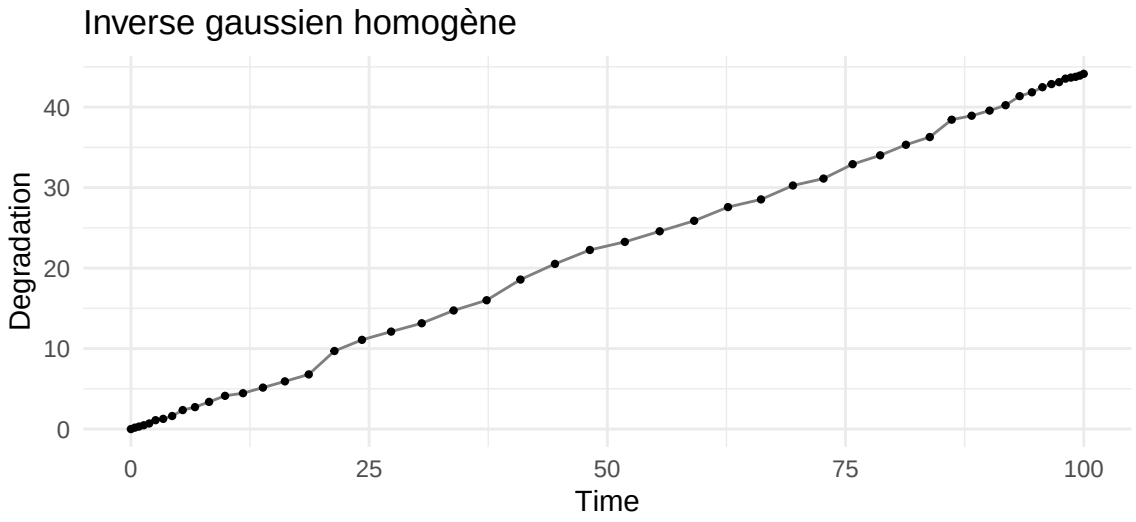
Modèle 2 : Wiener homogène

```
plot_traj(sim_wiener(t4, mu = 0.5, sigma = 1.5), "Wiener homogène")
```



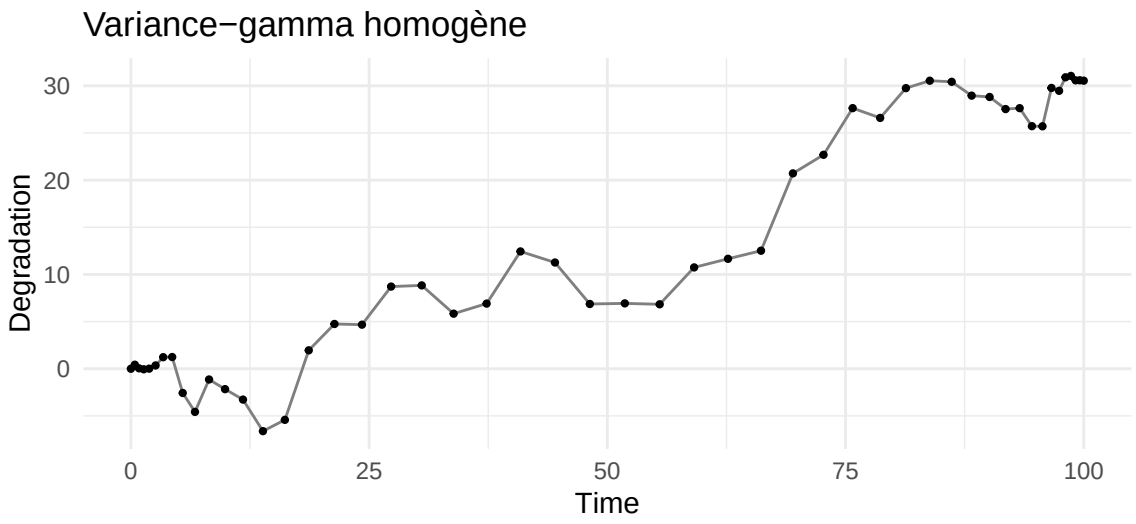
Modèle 3 : inverse gaussien

```
plot_traj(sim_ig(t4, mu = 0.5, eta = 2), "Inverse gaussien homogène")
```



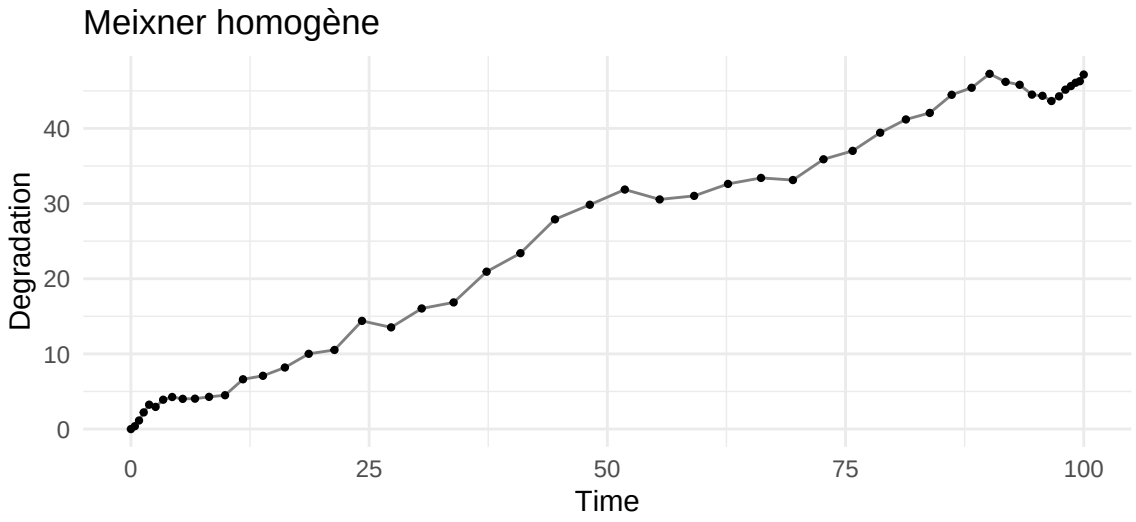
Modèle 4 : variance-gamma

```
plot_traj(sim_vg(t4, theta = 0.5, sigma = 1.5, nu = 2), "Variance-gamma homogène")
```



Modèle 5 : Meixner

```
plot_traj(sim_meixner(t4, a = 1, b = 0.3, d = 1.5, m = 0.2), "Meixner homogène")
```

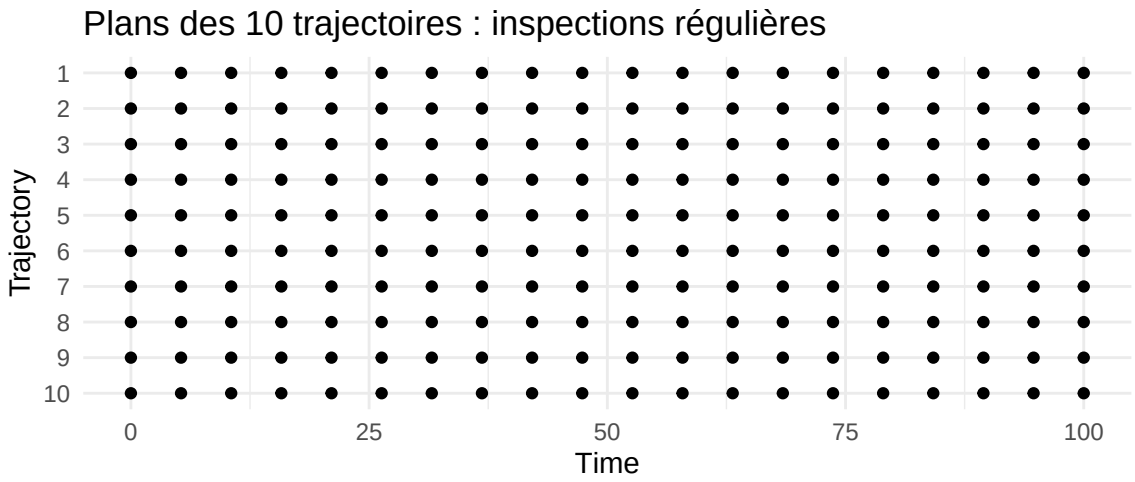


II - Plusieurs trajectoires

10 trajectoires, chacune inspectée 20 fois entre 0 et 100.

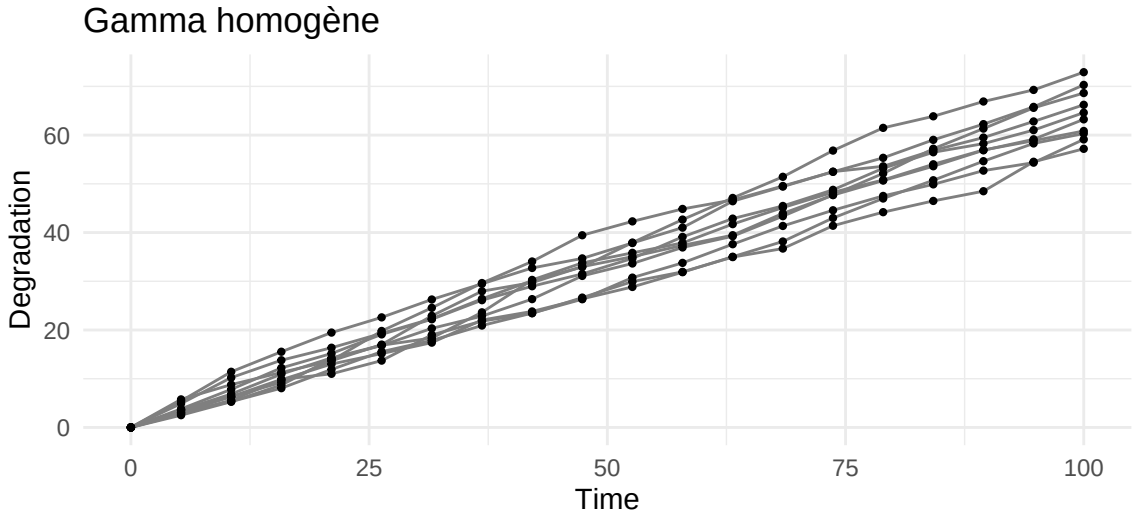
Cas 1 : inspections régulières

```
set.seed(5)
p1 <- inspection_times("regular", n = 19, n_traj = 10, tmax = 100)
plot_plan(p1, "Plans des 10 trajectoires : inspections régulières")
```



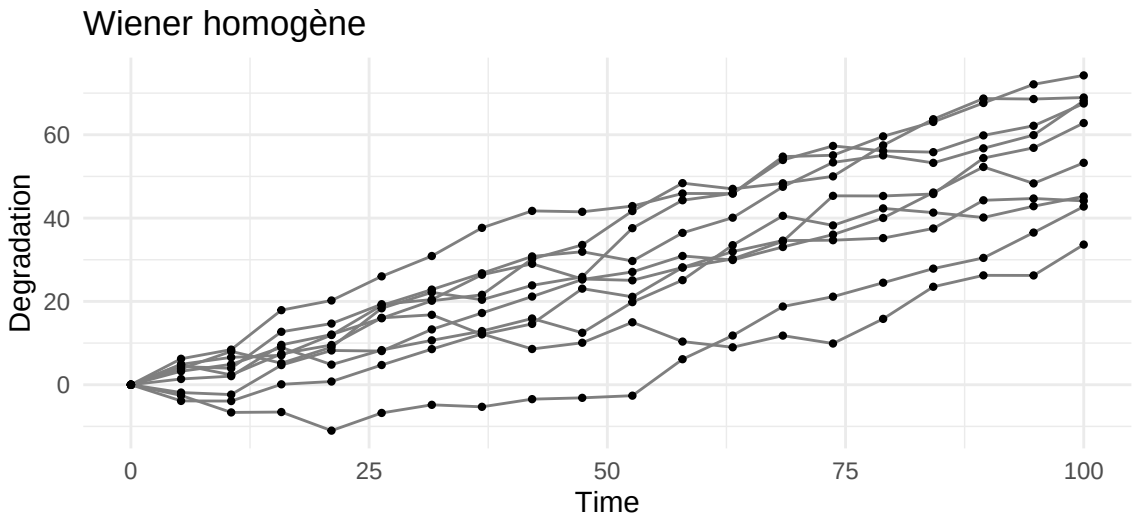
Modèle 1 : gamma homogène

```
plot_traj(sim_gamma(p1, a = 2, b = 3), "Gamma homogène")
```



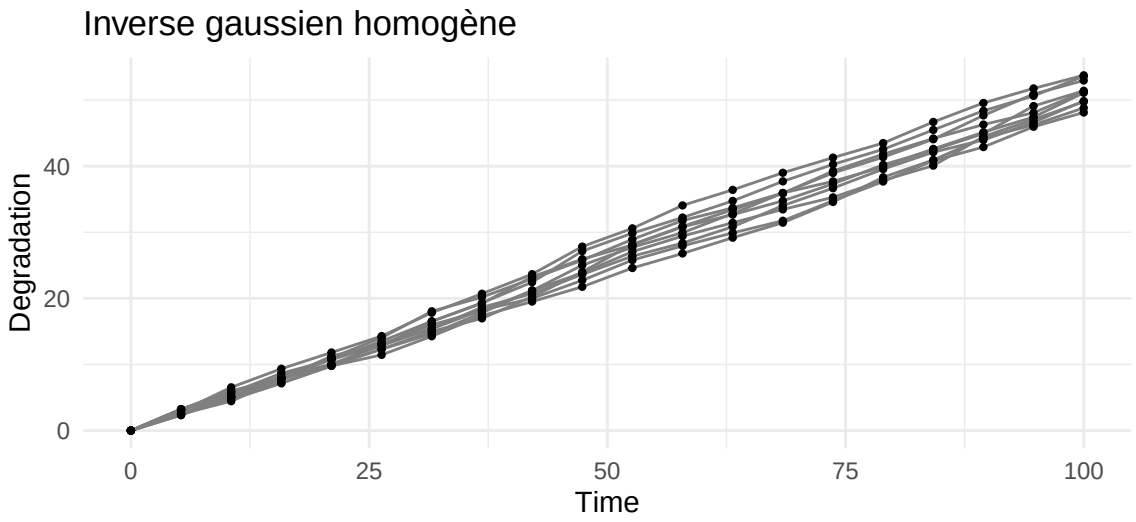
Modèle 2 : Wiener homogène

```
plot_traj(sim_wiener(p1, mu = 0.5, sigma = 1.5), "Wiener homogène")
```



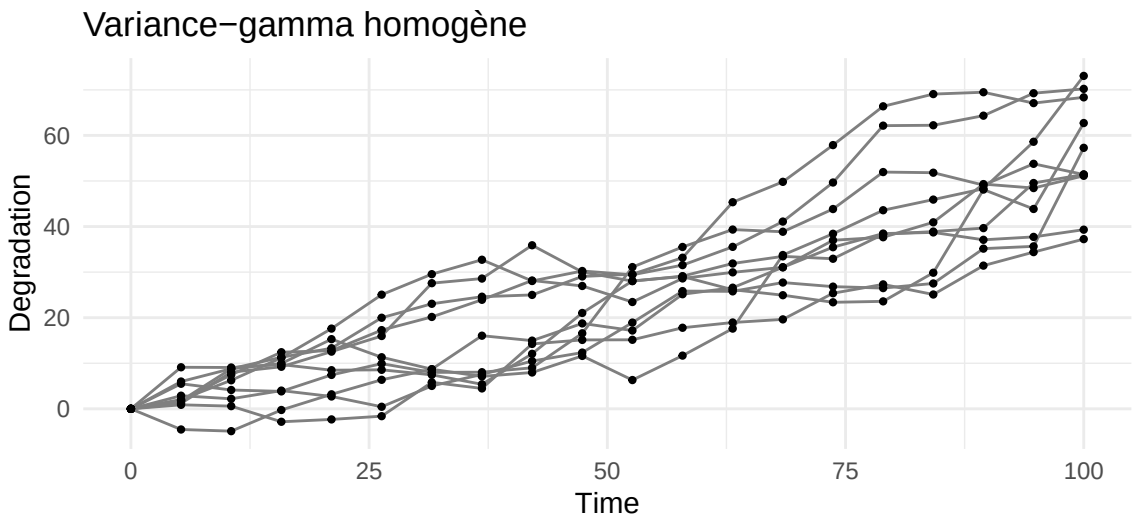
Modèle 3 : inverse gaussien

```
plot_traj(sim_ig(p1, mu = 0.5, eta = 2), "Inverse gaussien homogène")
```



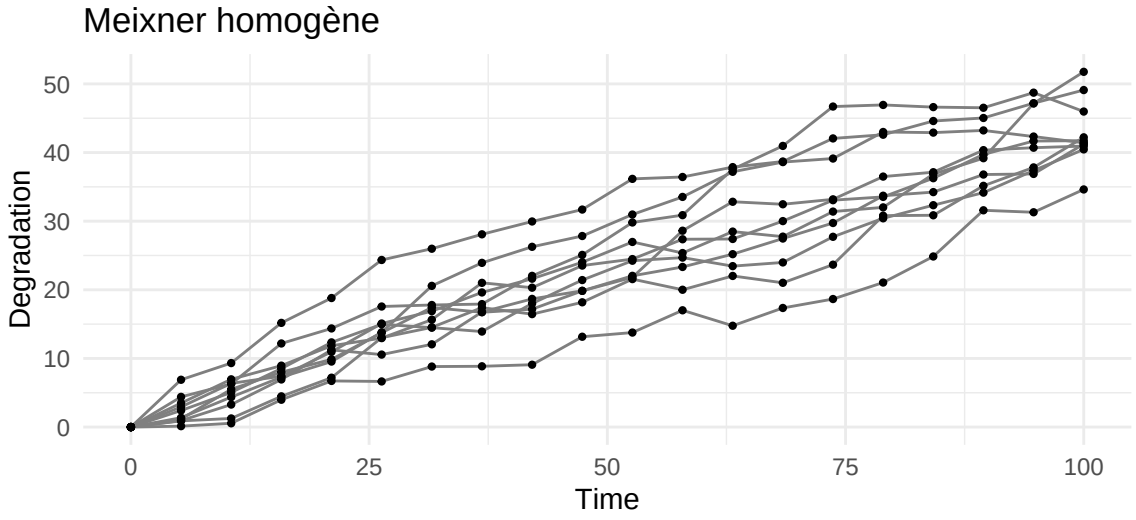
Modèle 4 : variance-gamma

```
plot_traj(sim_vg(p1, theta = 0.5, sigma = 1.5, nu = 2), "Variance-gamma homogène")
```



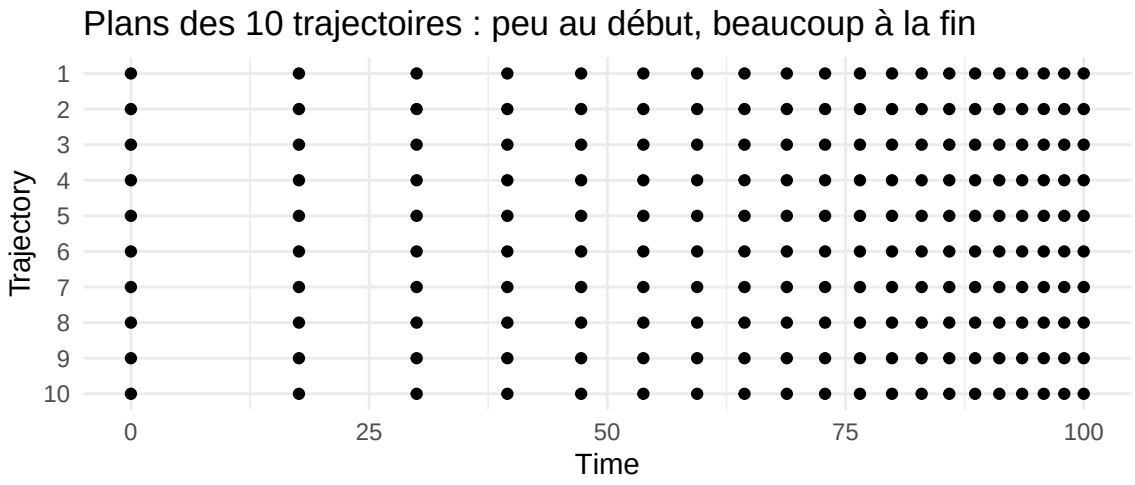
Modèle 5 : Meixner

```
plot_traj(sim_meixner(p1, a = 1, b = 0.3, d = 1.5, m = 0.2), "Meixner homogène")
```



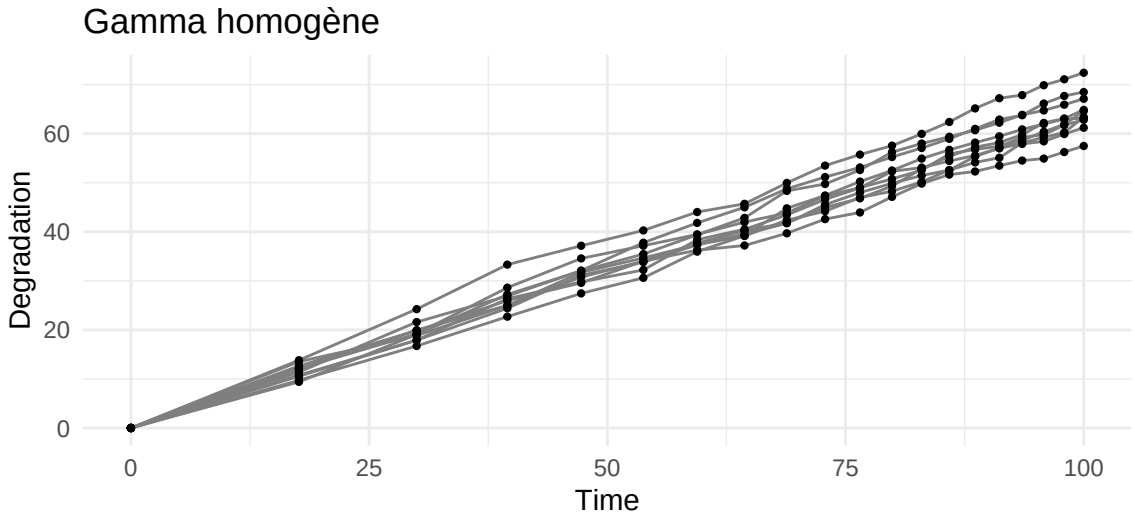
Cas 2 : inspections croissantes

```
set.seed(6)
p2 <- inspection_times("increasing", n = 19, n_traj = 10, tmax = 100)
plot_plan(p2, "Plans des 10 trajectoires : peu au début, beaucoup à la fin")
```



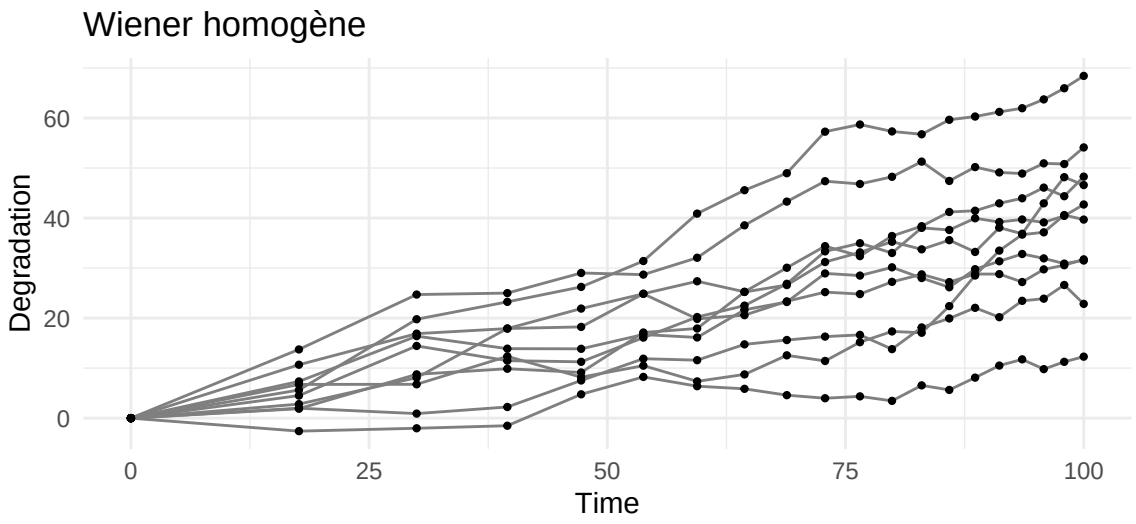
Modèle 1 : gamma homogène

```
plot_traj(sim_gamma(p2, a = 2, b = 3), "Gamma homogène")
```



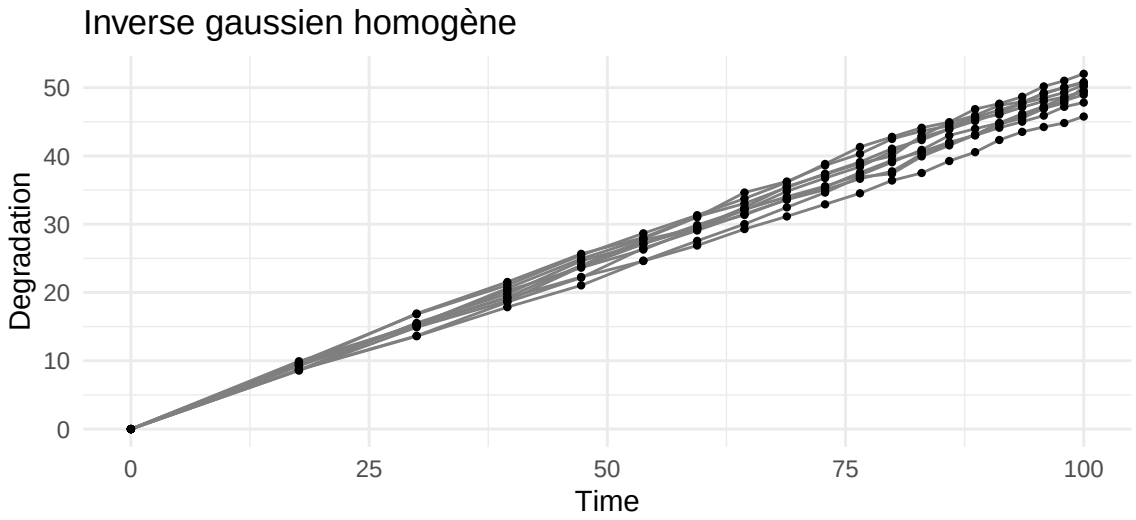
Modèle 2 : Wiener homogène

```
plot_traj(sim_wiener(p2, mu = 0.5, sigma = 1.5), "Wiener homogène")
```



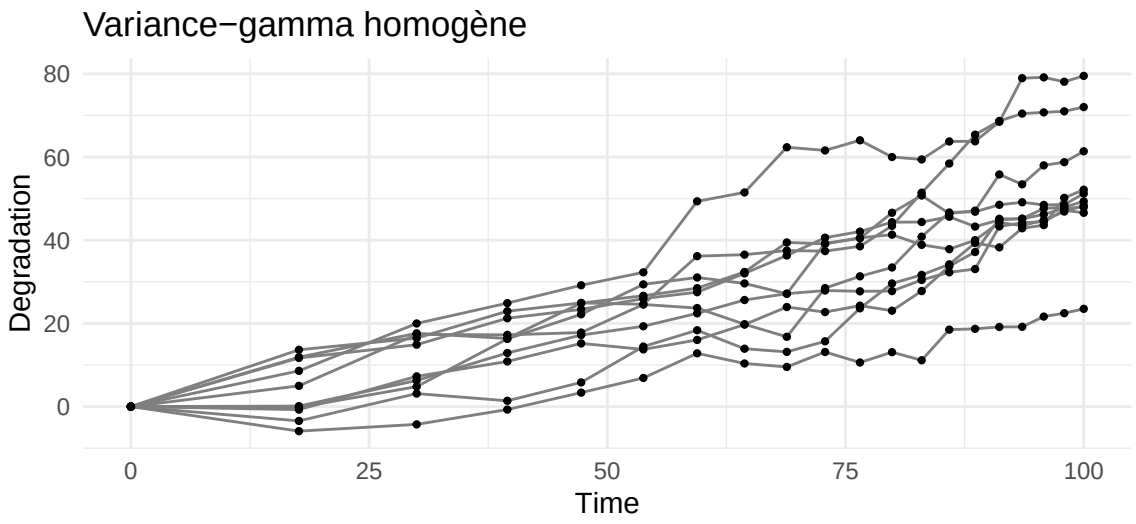
Modèle 3 : inverse gaussien

```
plot_traj(sim_ig(p2, mu = 0.5, eta = 2), "Inverse gaussien homogène")
```



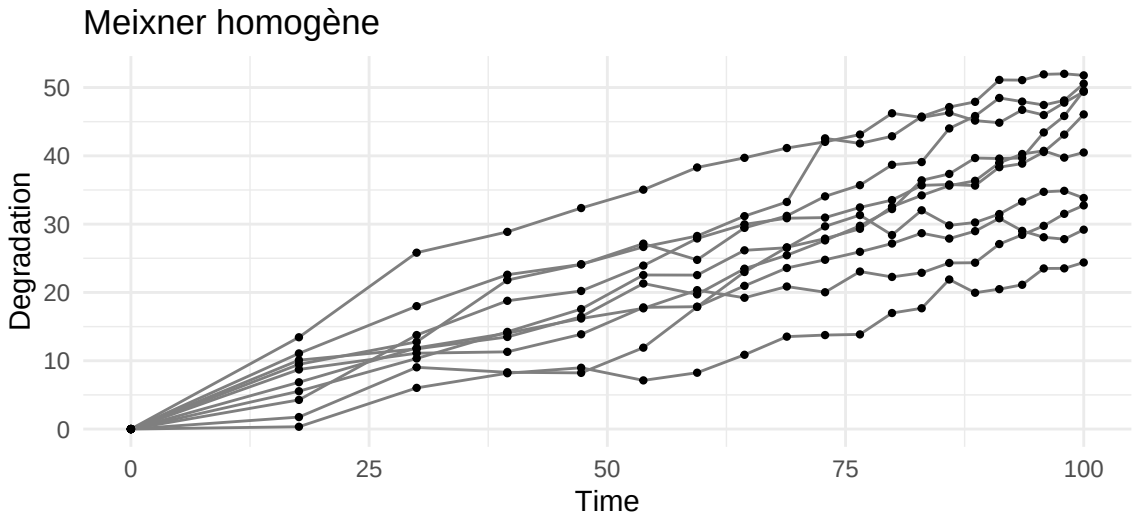
Modèle 4 : variance-gamma

```
plot_traj(sim_vg(p2, theta = 0.5, sigma = 1.5, nu = 2), "Variance-gamma homogène")
```



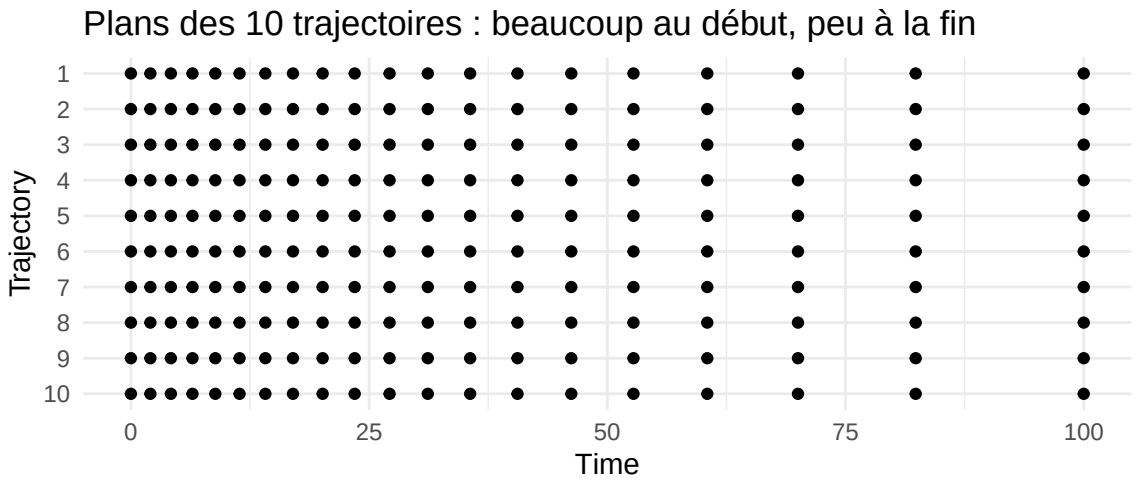
Modèle 5 : Meixner

```
plot_traj(sim_meixner(p2, a = 1, b = 0.3, d = 1.5, m = 0.2), "Meixner homogène")
```



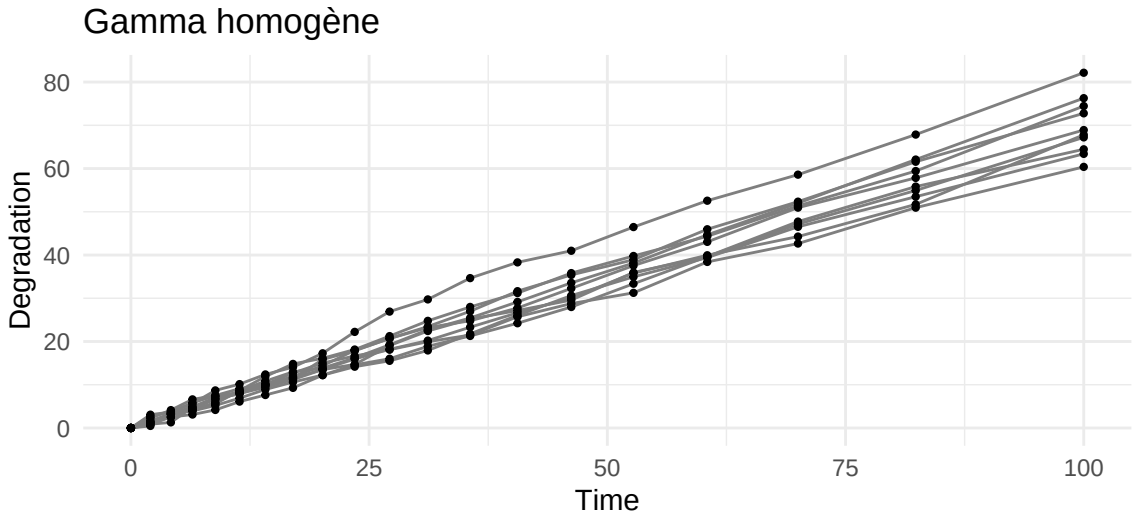
Cas 3 : inspections décroissantes

```
set.seed(7)
p3 <- inspection_times("decreasing", n = 19, n_traj = 10, tmax = 100)
plot_plan(p3, "Plans des 10 trajectoires : beaucoup au début, peu à la fin")
```



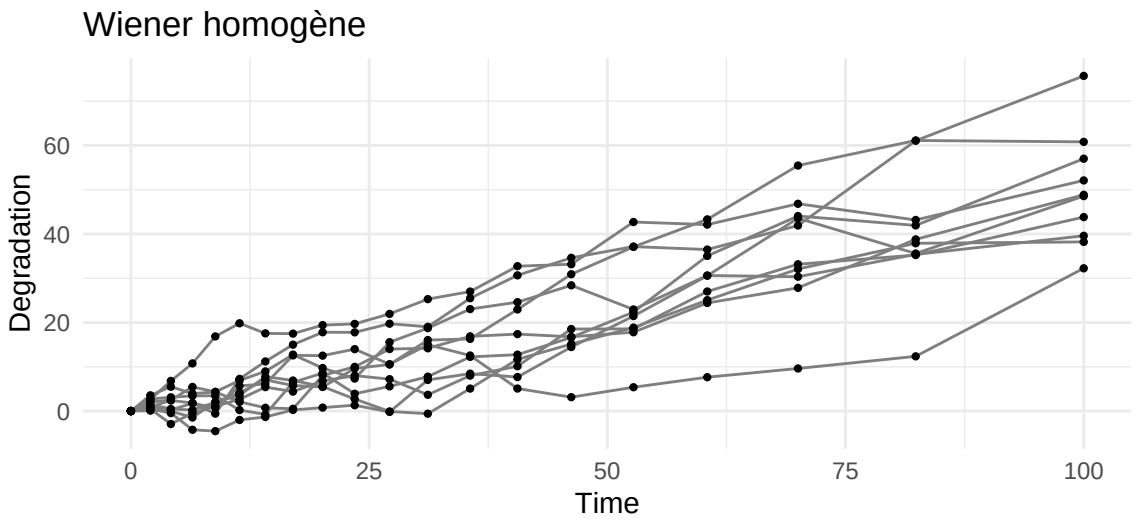
Modèle 1 : gamma homogène

```
plot_traj(sim_gamma(p3, a = 2, b = 3), "Gamma homogène")
```



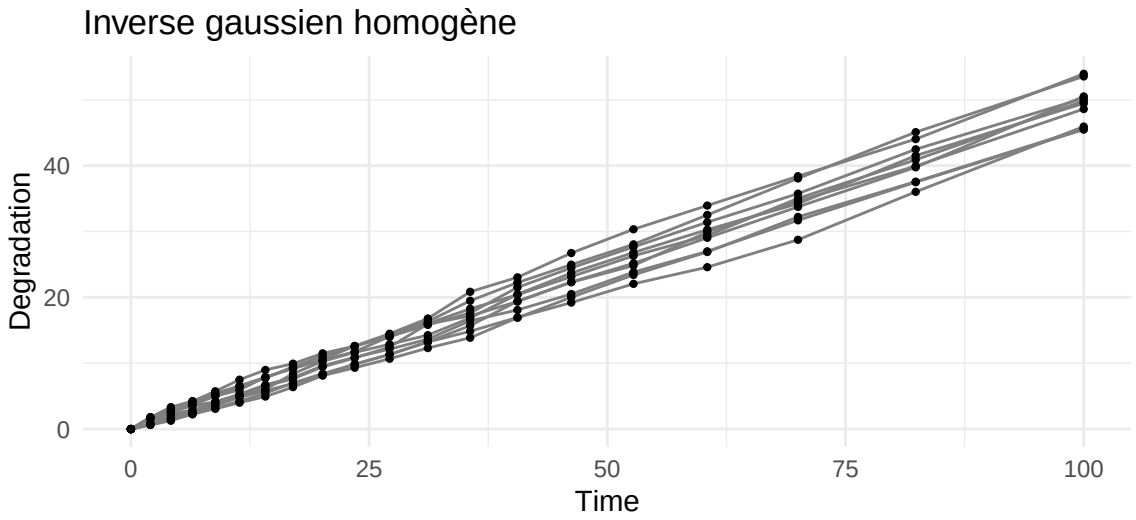
Modèle 2 : Wiener homogène

```
plot_traj(sim_wiener(p3, mu = 0.5, sigma = 1.5), "Wiener homogène")
```



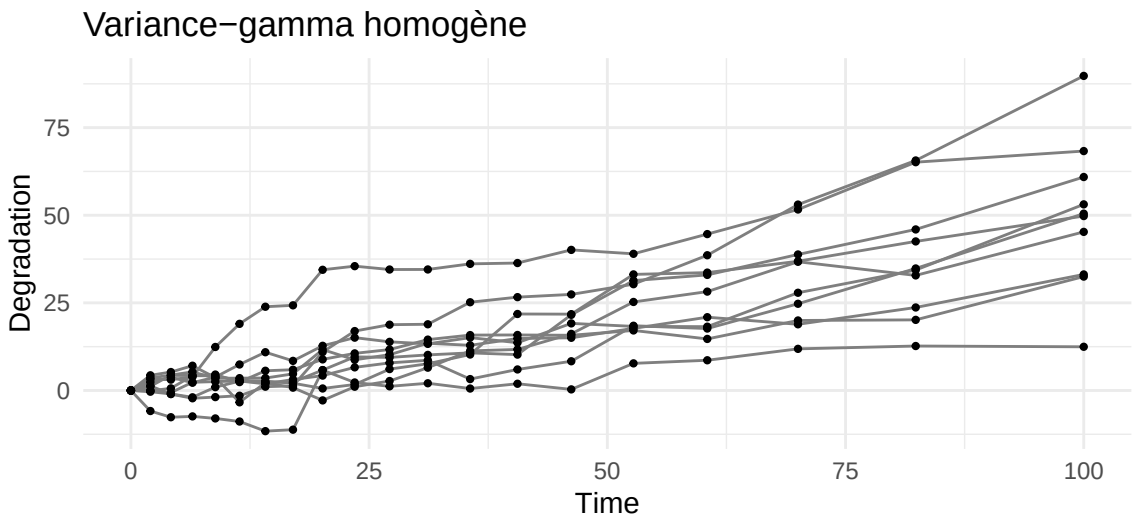
Modèle 3 : inverse gaussien

```
plot_traj(sim_ig(p3, mu = 0.5, eta = 2), "Inverse gaussien homogène")
```



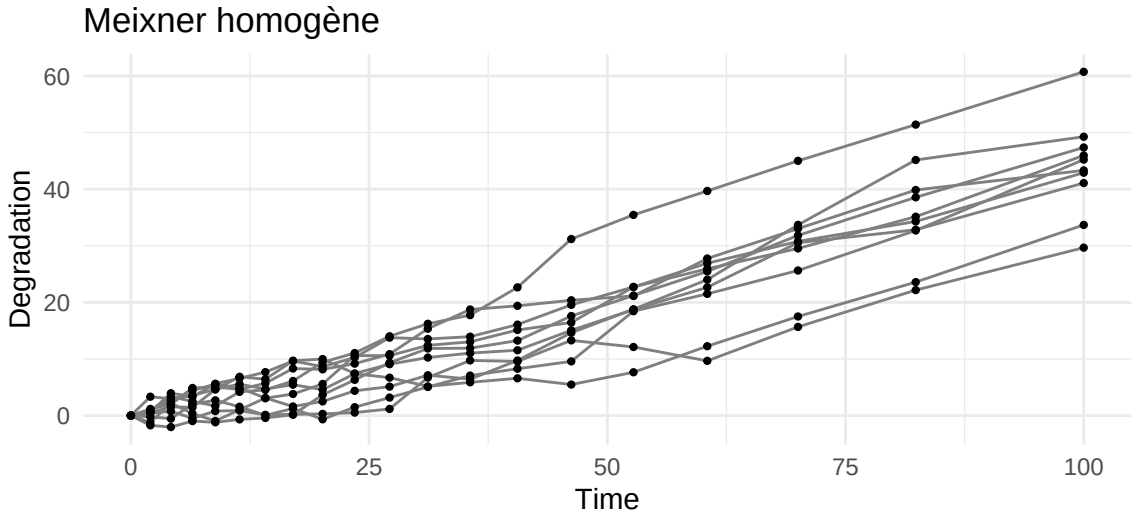
Modèle 4 : variance-gamma

```
plot_traj(sim_vg(p3, theta = 0.5, sigma = 1.5, nu = 2), "Variance-gamma homogène")
```



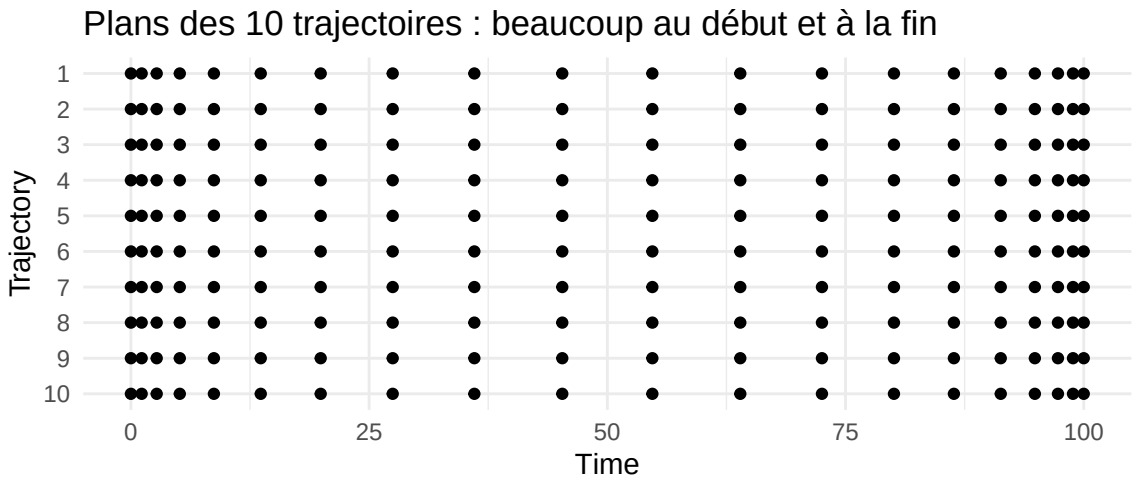
Modèle 5 : Meixner

```
plot_traj(sim_meixner(p3, a = 1, b = 0.3, d = 1.5, m = 0.2), "Meixner homogène")
```



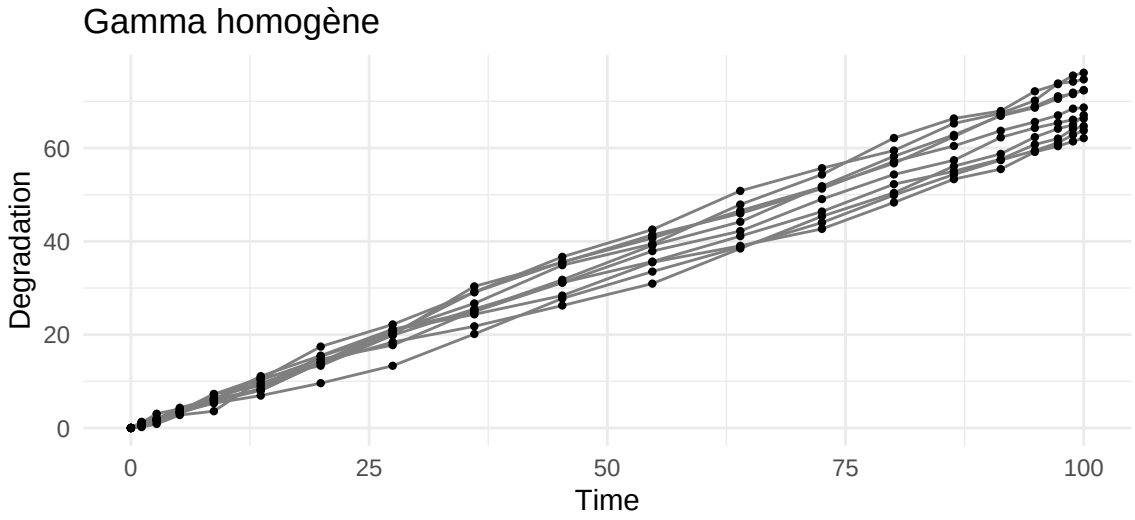
Cas 4 : inspections mixtes

```
set.seed(8)
p4 <- inspection_times("dec-inc", n = 19, n_traj = 10, tmax = 100)
plot_plan(p4, "Plans des 10 trajectoires : beaucoup au début et à la fin")
```



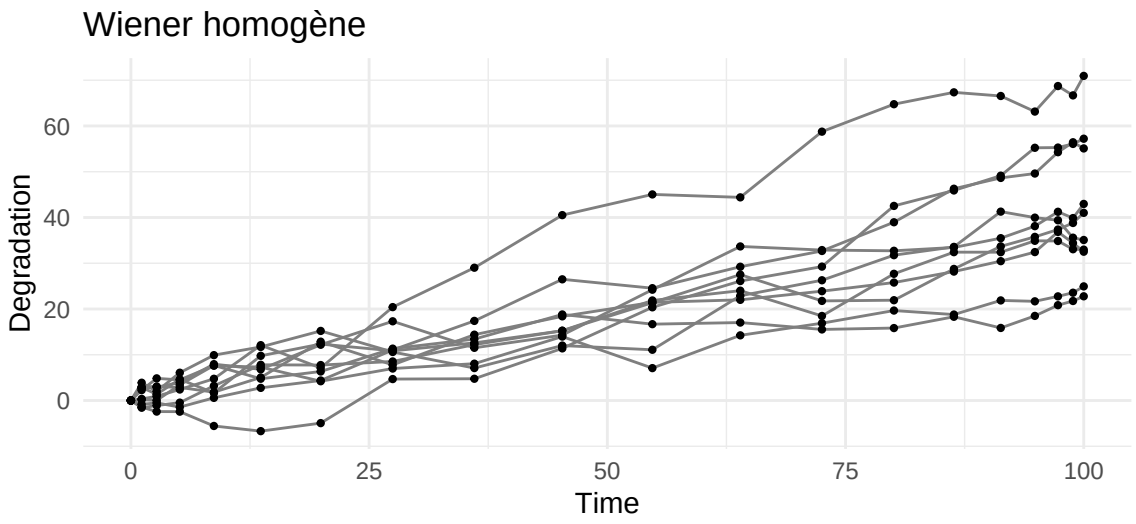
Modèle 1 : gamma homogène

```
plot_traj(sim_gamma(p4, a = 2, b = 3), "Gamma homogène")
```



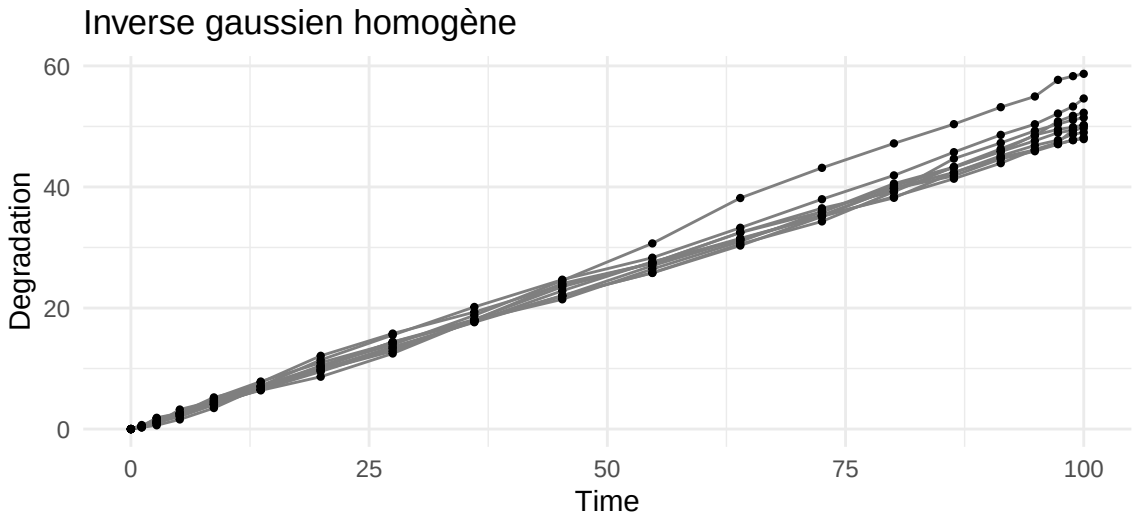
Modèle 2 : Wiener homogène

```
plot_traj(sim_wiener(p4, mu = 0.5, sigma = 1.5), "Wiener homogène")
```



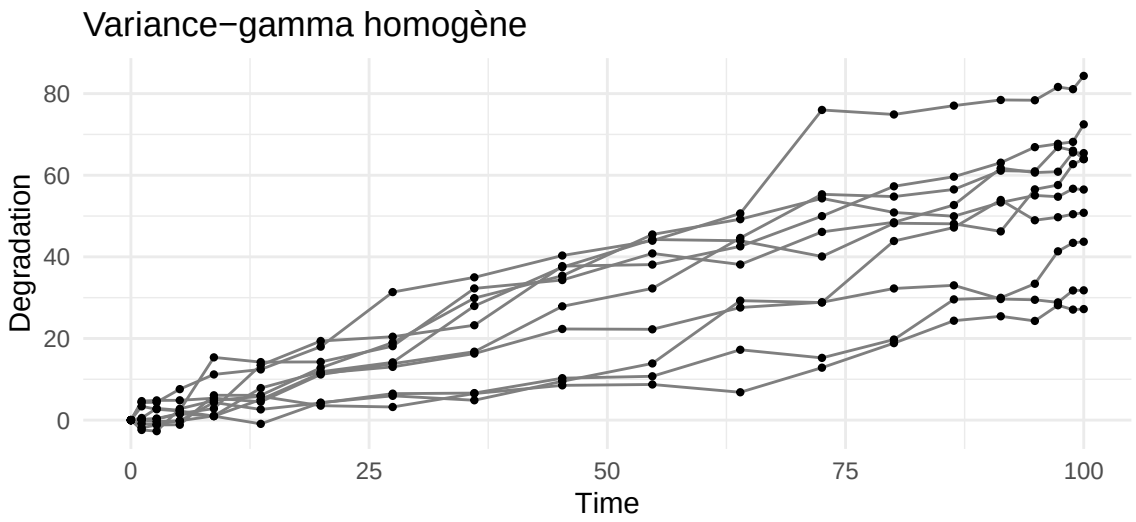
Modèle 3 : inverse gaussien

```
plot_traj(sim_ig(p4, mu = 0.5, eta = 2), "Inverse gaussien homogène")
```



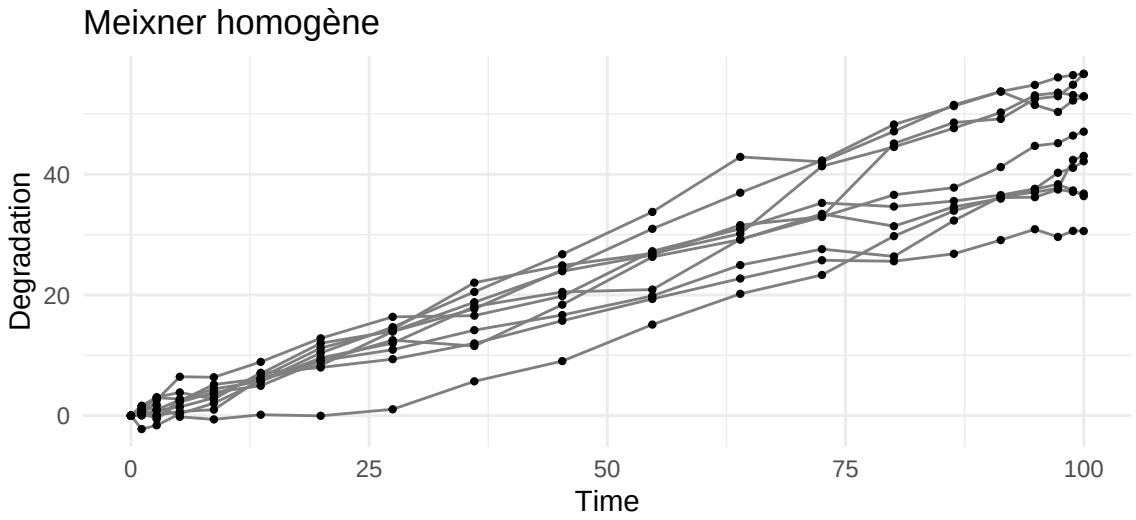
Modèle 4 : variance-gamma

```
plot_traj(sim_vg(p4, theta = 0.5, sigma = 1.5, nu = 2), "Variance-gamma homogène")
```



Modèle 5 : Meixner

```
plot_traj(sim_meixner(p4, a = 1, b = 0.3, d = 1.5, m = 0.2), "Meixner homogène")
```



III - Récupérer les données simulées

Le résultat d'une fonction `sim_` s'affiche directement : modèle, paramètres et premières valeurs.

```
set.seed(9)
plan <- inspection_times("increasing", n = 9, n_traj = 3, tmax = 100)
x <- sim_gamma(plan, a = 2, b = 3)
x
```

```
## 3 simulated trajectory(ies) of the gamma process
## parameters: a = 2, b = 3
##
##   id        t        X
## 1  1  0.00000  0.00000
## 2  1 31.16126 18.64680
## 3  1 48.79576 28.74181
## 4  1 61.15101 36.36308
## 5  1 70.67013 46.49263
## 6  1 78.41514 52.92328
```

`x[[k]]` donne la k-ième trajectoire, sous la forme d'un `data.frame(t, X)` :

```
x[[2]]
```

```
##            t        X
## 1    0.00000  0.00000
## 2   31.16126 18.46611
## 3   48.79576 32.71991
## 4   61.15101 42.81629
## 5   70.67013 51.45697
## 6   78.41514 57.38427
## 7   84.94457 62.29249
## 8   90.58875 64.95217
## 9   95.55929 69.11958
## 10 100.00000 71.28436
```

`traj_df()` rassemble toutes les trajectoires dans un seul tableau, avec une colonne `id` :

```
d <- traj_df(x)
head(d)
```

```
##   id        t        X
## 1  1  0.00000  0.00000
## 2  1 31.16126 18.64680
## 3  1 48.79576 28.74181
## 4  1 61.15101 36.36308
## 5  1 70.67013 46.49263
## 6  1 78.41514 52.92328
```

```
d[d$t == 100, ]
```

```
##    id   t        X
## 10  1 100 67.38996
## 20  2 100 71.28436
## 30  3 100 68.20323
```

Ce tableau peut être enregistré, par exemple dans un fichier CSV :

```r
write.csv(d, "trajectoires_gamma.csv", row.names = FALSE)
```

Le graphique renvoyé par `plot_traj()` se modifie avec les fonctions de `ggplot2`, par exemple pour ajouter un seuil de défaillance :

```r
plot_traj(x, "Gamma homogène : 3 trajectoires") +
  geom_hline(yintercept = 60, linetype = "dashed", colour = "red") +
  labs(x = "Temps (heures)", y = "Dégradation")
```

